

Proton therapy system

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1. Beam production

We can use the information given concerning the output proton energy to compute the relativistic velocity of output proton using the formula:

$$v_{rel} = c * \sqrt{1 - (1 + \frac{E}{m})^{-2}}$$

where E is the output proton energy [Mev], and m is the proton mass [MeV/c²]. We can then calculate the Lorentz factor using the formula:

$$\gamma = \frac{1}{\sqrt{1 - (\frac{v_{rel}}{c})^2}}$$

and we can use it to calculate the relative mass of the proton at the output: $m_{rel} = m * \gamma$.

Since we know the mass, velocity, charge and radius of a particle, we can calculate the magnetic field required to keep the particle in the trajectory of radius R_{max} :

$$B_{extr} = \frac{m_{rel} * v_{rel}}{q * R_{max}}$$

where B_{extr} is the magnetic field at extraction.

Since we know the magnetic field and mass of the particle at a given point in the cyclotron (extraction), we can calculate the cyclotron frequency:

$$f_{cyclotron} = \frac{q * B_{extr}}{2 * \pi * m_{rel}}$$

By taking its reciprocal, we find the period of revolution, i.e. the time it takes the particle to complete one revolution. Knowing the total acceleration time, we can find how many revolutions the particle makes before exiting the cyclotron.

By inverting the formula: $E = q * 2 * n_{dees} * n_{turn} * V_{gap}$ we can calculate the high voltage amplitude to be imposed in each gap.

Whereas, since there are 2 gaps, the frequency of high voltage is given by $f_{cyclotron}$ divided by two.

Considering that at injection the mass is stationary, the mass and charge of the particle corresponds to that of a stationary proton. We can calculate the magnetic field at the center of the cyclotron, since we have calculated the cyclotron frequency, which is constant throughout the cyclotron.

In conclusion, the parameters to fulfill the objectives are:

1. Magnetic field at injection: 1.971 T
2. Magnetic field at extraction: 2.391 T
3. Cyclotron frequency: 30.036 MHz
4. Frequency of high voltage: 15.018 MHz
5. Number of revolutions: 27032
6. Required high voltage amplitude per gap: 1.85 kV

2. Beam transport

Firstly, we defined all the necessary variables, including the initial beam size, divergence in both directions, and the vacuum tube radius. Using the *initial_matrix(N)* function, we create the beam size matrix, a matrix with 4 columns and N rows. For each row, we initialize the position and divergence of a single particle, sampling these values from a normal distribution with zero mean and a standard deviation specified by the delivery parameters.

Next, we define two functions that will be useful in the subsequent lines of code:

drift_matrix(d), which returns a 4x4 drift matrix given the drift distance *d*.

focus_matrix(f), which returns a 4x4 quadrupole matrix.

Depending on the sign of *f*, the quadrupole focuses the beam in one direction and defocuses it in the other.

- If $f > 0$, the beam is focused on the y-direction and defocused in the x-direction.
- If $f < 0$, the beam is focused on the x-direction and defocused in the y-direction.

We can think of dividing the beam transport into 8 steps, alternately representing the beam after a drift distance or after passing through a quadrupole. To implement these calculations, we define the function *propagate_stepwise(initial_beam, d1, d2, d3, d4, f1, f2, f3, f4)* which, given the initial beam size matrix (*initial_beam*), all the distances (*d*), and all the focal lengths (*f*), returns a vector of 8 beam size matrices. Each matrix is obtained by multiplying the current beam size matrix by the corresponding drift or quadrupole matrix, created using the functions defined above. Thanks to this function, we can determine the beam size matrix at each step of the beam transport, as the positioning of the quadrupoles along the vacuum tube and their focal lengths vary.

Now, to find the best combination of the quadrupole distances and their focal lengths that satisfy all objectives and constraints, we can implement a cost function and minimize it. We define the cost function, which, given all the parameters (distances *d* and focal lengths *f*), computes the beam size matrices for each step (using the *propagate_stepwise* function) and returns the cost as a combination of the symmetry penalty (which ensures that the beam must be symmetrical at the isocenter at the end of the vacuum tube) and the final size penalty (which ensures that the beam size (1 sigma) must not exceed 3 mm at the isocenter).

We decided to include the other two constraints (“distance between the output of the cyclotron and the isocenter where the patients will be treated is 7 m” and “the full beam size must always stay smaller than the radius of the vacuum tube”) directly within the Python *minimize* function. This approach ensures that these important constraints are handled appropriately, while keeping the cost function simpler and more manageable. To achieve this, we define two functions to calculate the constraints:

- *constraint_total_length*, which, given the parameters, returns the difference between the available distance to direct the beam and the sum of the distances between all the quadrupoles. Since we want this difference to be 0, we will define this constraint, inside the *minimize* function, with 'type': 'eq'.

- `constraint_tube`, which, given the parameters and using the `propagate_stepwise` function, returns a vector of 16 values representing the difference between the radius of the vacuum tube and the full beam size ($3 * \sigma_y$ and $3 * \sigma_x$) at each step, ensuring that the beam never touches the wall of the vacuum tube. Since we want all these differences to be < 0 , we will define this constraint, inside the `minimize` function, with 'type': 'ineq'.

To recap, we aim to find the optimal parameters (d and f) that minimize the cost function, while simultaneously satisfying the constraints defined in the constraints variable.

In order to make the problem physically realistic and to make the optimization more effective, we impose bounds on the parameters. We also define an initial guess where the signs of the focal lengths alternate.

To ensure that the final beam dimensions in both the x and y directions are sufficiently close, we use a while loop that repeats the optimization until the symmetry condition is satisfied.

Inside the loop, we compute the result of the `minimize` function, which takes as input: `cost_function`, `initial_guess`, `bounds`, and the previously defined constraints. The optimized parameters are saved, and the standard deviation of the final beam is calculated to check for symmetry.

Finally, we printed out the results of the optimized parameters and the size of the final beam at the isocenter (`sigma_x_final` and `sigma_y_final`), to ensure that it is symmetrical and no larger than 3mm.

By initializing an initial beam with 5000 particles and running the code multiple times, we observe that the parameters optimized by the cost function converge to the values shown below:

$d1 = 1.52 \text{ m}$, $d2 = 1.95 \text{ m}$, $d3 = 1.83 \text{ m}$, $d4 = 1.70 \text{ m}$

$f1 = 2.54 \text{ m}$, $f2 = -1.66 \text{ m}$, $f3 = 1.45 \text{ m}$, $f4 = -2.06 \text{ m}$

with the final beam size at the isocenter: 2.99 mm (X), 2.99 mm (Y).

We can observe that the quadrupoles are evenly distributed along the vacuum tube, and that the initial assumption of alternating the signs of the focal lengths is maintained. This alternating arrangement allows the beam to be defocused by one quadrupole and then refocused by the next, for each direction. Furthermore, we can state that from the dimensions of the final beam at the isocenter, the beam at the isocenter meets the required conditions.

Using the parameters found we can plot the beam for each of the 8 steps, what we obtain is shown below:

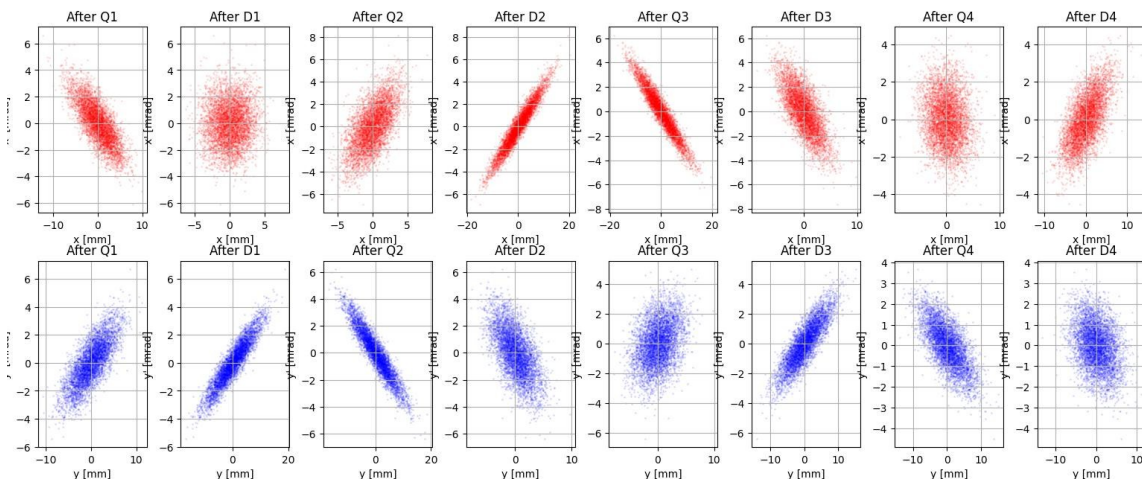


Figure 1: representation of the beam in phase-space at the different locations of interest along the beamline

3. Beam delivery

First, we initialized all the target and pencil beam parameters provided by the delivery system.

To create the Gaussian spot template, we generated an array centered at zero to ensure symmetry in the distribution. The *gauss1D* variable stores a one-dimensional Gaussian distribution, which we then use to construct a two-dimensional Gaussian distribution representing our spot.

We initialize the dose grid and set the starting position of the beam within it.

Using two loops, we move in the x and y directions within the dose grid. After checking that the current position of the spot is still inside the dose grid, we add the spot to it.

To normalize the total dose so that the average dose in the target equals 1 Gy, we create a mask (with the same dimensions as the dose grid) that contains 1 where the target is present and 0 elsewhere. We use this mask to calculate the current average dose for each pixel within the target. Finally, we apply the dose normalization to ensure that the average dose in the target equals 1 Gy.

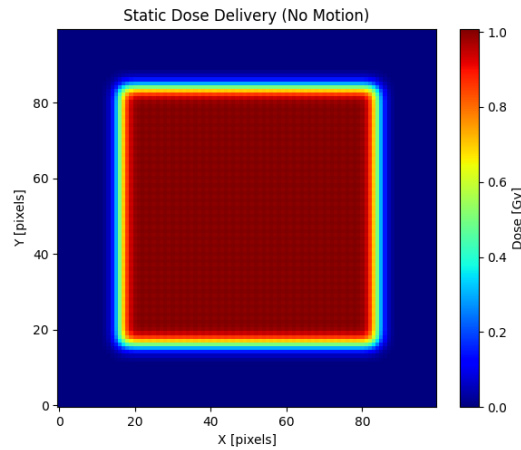


Figure 2: dose distribution without motion

Introduction of motion

To calculate the dose distribution in the case of horizontal periodic motion, we defined the parameters of the periodic triangle wave representing the motion.

By creating a function, we computed the wave for two full breathing cycles (8 seconds). Then, we created the *simulate_interplay* function to compute the dose delivery with motion, considering both cases: when the scanning lines are parallel to the motion and when they are orthogonal to the motion. This function is implemented using the same loops as those used for the calculation of the dose without motion, with the difference that the new position of the spot in the x and y directions within the dose grid is computed by considering the body displacement at each time instant. To do so, we calculate the time instant for the current iteration using the variable *spot_index* (which is updated for each iteration) and knowing that the time between two consecutive pencil beams is 22 ms. Knowing the current time instant, we can calculate the relative displacement of the body using the previously created function.

In the case where the scanning lines are parallel to the motion, we must update the x position of the spot inside the grid by subtracting the calculated displacement. In the case where the scanning lines

are orthogonal to the motion, we need to subtract the displacement when updating the y position. The rest of the loop is the same as the previous one.

Simulating the interplay in both cases, the resulting plots are as follows:

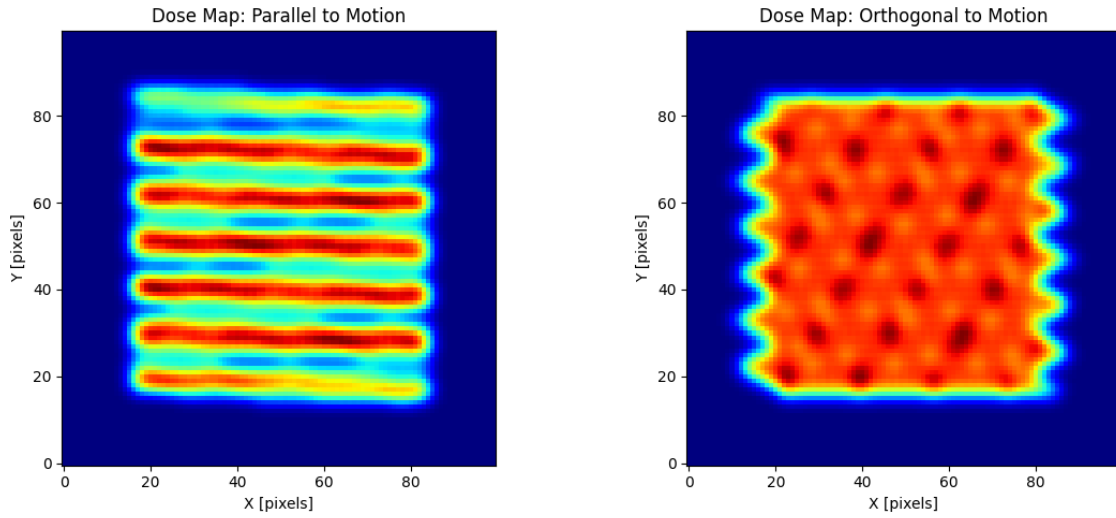


Figure 3: dose distribution with motion

Compute and display coverage statistics

To assess if at least 95% of the target receives >0.9 Gy, we create a function that, given the previously computed dose distribution (*dose_parallel* and *dose_orthogonal*) as input, returns the percentage of voxels that received a dose > 0.9 Gy. This function computes the number of pixels that received a dose > 0.9 Gy, and by dividing this number by the total number of voxels in the target, it calculates the percentage.

The results show that when the scan lines are parallel to the motion, only 54% of the target receives $\geq 90\%$ of the prescribed dose. In contrast, when the lines are orthogonal, the objective is fully achieved (98%). This big difference was already visible from figure 3

Interplay reduction strategy – Adaptive Spot Triggering

To reduce the interplay effect, we can consider firing the beam spot only when the tumor is aligned with it, within a certain tolerance. To do so, we create the *simulate_adaptive_spot_triggering* function, which, like the *simulate_interplay* function, computes the current displacement considering the spot index. But in this case, the spot is added to the grid dose only if the displacement is lower than the tolerance value provided as input to the function. We can implement this using a while loop that updates the time and calculates the current displacement until the condition is no longer satisfied. The beam spot is fired, taking into account the displacement in the x-axis. For the rest, the function is the same as above.

We choose several possible tolerances, and for each of them, we calculate the dose and the percentage of the target receiving at least 90% of the prescribed dose. From the results, we can see that the target is only reached with a tolerance of 2 mm or greater. The dose map using this strategy turns out to be:

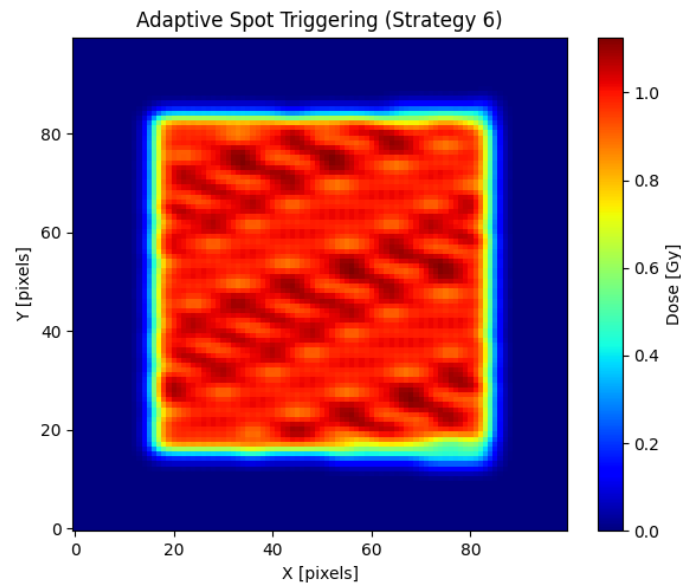


Figure 4: dose distribution with adaptive spot triggering

4. Treatment safety

The main risks associated with using adaptive spot triggering are:

Risk 1: Increased treatment time. The use of Adaptive Spot Triggering may increase treatment time, as the system may have to wait extended periods to align the tumour correctly before each spot. Lengthy treatment can be problematic, especially for patients who cannot remain immobile for long periods.

- Mitigation strategy: To avoid excessively long treatment sessions, a maximum time-out per spot could be implemented, limiting the time the system can wait before activating the treatment. This could be based on clinical parameters, such as tumour mobility, or on an algorithm to estimate the optimal time for alignment.

Risk 2: Incomplete target coverage. Another risk associated with Adaptive Spot Triggering is the possibility of some spots being missed frequently during treatment. This could lead to uneven dose distribution, with some areas of the target receiving less dose than prescribed.

-Mitigation strategy: To reduce the risk of incomplete coverage, rescanning strategies or “fallback delivery” methods can be combined to ensure that all target areas receive the intended dose.

Risk 3: Error or delay in monitoring movement. Accurate monitoring of tumour movement is essential to ensure that treatment is delivered at the right time. If there are errors or delays in identifying the location of the tumour, beam delivery may be timed incorrectly, with the risk of damaging surrounding tissues or not treating the tumour correctly.

- Mitigation strategy: One of the main solutions is the adoption of advanced motion monitoring systems, such as real-time dynamic imaging or highly accurate motion sensors. Finally, technical limitations of beam delivery systems must be considered to ensure clinical feasibility.

Final considerations: Adaptive Spot Triggering offers a highly accurate approach to treating mobile tumours, but its safe implementation requires a combination of advanced technologies, risk mitigation strategies and strict quality protocols. It is crucial to ensure that the motion monitoring system is reliable. In summary, a combination of accurate monitoring, recovery strategies and time limits must be adopted to ensure treatment safety.